

Received 27 May 2026, accepted 18 June 2026, date of publication 25 June 2026, date of current version 30 June 2026.

Digital Object Identifier 10.1109/ACCESS.2026.3707232

RESEARCH ARTICLE

YOLOv8-PaveDet: Lightweight UAV Pavement Distress Detection With Context-Enhanced Features and Class-Aware Reweighting

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This work was supported in part by the 2025 Higher Education Institution Scientific Research Program of Anhui Provincial Department of Education under Grant 2025AHGXSK30415, in part by the 2024 University-Level Scientific Research Program of Tongling University under Grant 2024tlxyskZD07, and in part by the 2024 Talent Research Start-Up Fund Project of Tongling University under Grant 2024tlxyc005.

ABSTRACT High-resolution pavement imagery collected by uncrewed aerial vehicles (UAVs) enables scalable, non-intrusive road distress inspection. Combined with lightweight detectors, it supports flexible automated detection of surface distress across large road networks. However, accurate detection remains challenging because thin cracks are visually subtle and easily fragmented, scale variation across UAV altitudes is pronounced, and safety-critical categories such as potholes are severely underrepresented. Existing approaches address these challenges only in isolation: context-enhanced Transformer architectures achieve strong feature modeling but are computationally prohibitive for high-resolution UAV tiles; lightweight detectors preserve efficiency but sacrifice contextual aggregation; and existing long-tail loss functions are designed for generic benchmarks without accounting for the joint difficulty of thin-structure localization and severe class imbalance in UAV pavement scenes. To address these limitations, we propose YOLOv8-PaveDet, a task-oriented extension of YOLOv8n with three coordinated components: GhostConv for efficient backbone computation, ConvFormer-enhanced C2f blocks for long-range spatial aggregation over elongated crack structures, and a lightweight shared-channel detection head (LSCD) for scale-consistent multi-level prediction. A hybrid training loss combining IoU-aware SlideLoss with class-weighted binary cross-entropy provides explicit minority-class rebalancing. Experiments on UAV-PDD show that the architectural modifications raise mAP@0.5 from 62.9% to 73.4% and mAP@0.5:0.95 from 32.4% to 42.5%, while reducing parameters from 3.01M to 1.83M and GFLOPs from 4.10 to 2.78. Adding the hybrid loss further improves mAP@0.5 to 81.68% (Aggressive) and mAP@0.5:0.95 to 45.54% (Conservative), with *Pothole* AP@0.5 rising from 39.70% to 75.81%. These results demonstrate that coordinated lightweight design, combining contextual aggregation with explicit class rebalancing, achieves a stronger accuracy–efficiency balance for UAV pavement inspection; the resulting detections can serve as upstream sensing inputs for public infrastructure management.

INDEX TERMS UAV pavement inspection, pavement distress detection, vision-based sensing, lightweight detector, long-tailed learning, crack detection, road monitoring.

I. INTRODUCTION

A. UAV-BASED PAVEMENT INSPECTION

Pavement distress inspection underpins road safety management, maintenance scheduling, and longitudinal

The associate editor coordinating the review of this manuscript and approving it for publication was Gerardo Di Martino.

condition monitoring. Conventional survey methods, manual inspections and vehicle-mounted systems, remain widely used, yet both approaches are labor-intensive and difficult to update frequently, especially when subtle distress such as hairline cracks must be recorded consistently [1], [2].

Uncrewed aerial vehicles (UAVs) offer a complementary sensing platform because they can cover road

corridors quickly, acquire high-resolution imagery, and revisit the same area with flexible deployment in urban and highway environments [3]. Pairing UAV imagery with deep learning enables automated distress detection and produces structured outputs compatible with downstream inspection workflows [4] including georeferenced crack maps and quantitative measurements derived through image stitching and segmentation [3].

B. TECHNICAL CHALLENGES AND RESEARCH GAPS

Reliable automated distress detection in UAV road scenes remains technically difficult. Pavement cracks are thin, elongated, and visually subtle, with poor spatial continuity and low contrast against heterogeneous pavement textures [2]. UAV imagery compounds these difficulties through non-uniform illumination, shadows from trees and buildings, specular highlights, as well as motion blur and partial occlusions. These nuisance factors can induce both false positives (textures or markings mistaken as distress) and false negatives (missed faint or fragmented patterns) [3].

A central bottleneck is the tension between preserving spatial detail and maintaining computational feasibility. UAV images are typically high-resolution; retaining hairline cracks requires fine spatial sampling. Mainstream detectors, however, rely on aggressive downsampling in the backbone and neck to gain receptive field and efficiency, which can attenuate or erase micro-structures that are critical for thin-distress recognition [2]. Processing full-resolution frames end-to-end is computationally expensive and often incompatible with memory and latency constraints, particularly in resource-constrained field deployment where on-site inference is preferred to reduce bandwidth and accelerate inspection feedback [5].

These constraints expose three gaps beyond incremental tuning of standard detectors. Feature extraction must reduce redundant computation over texture-heavy road regions while preserving crack-sensitive features. Context aggregation must connect long crack patterns without the quadratic cost of full self-attention on high-resolution UAV tiles. The training loss must rebalance rare safety-critical categories without weakening the learning signal for frequent defects.

Existing methods address these requirements only partly. Lightweight detectors such as YOLOv8n achieve the efficiency required for field deployment but provide limited context aggregation for thin structures, producing fragmented detections under cluttered backgrounds. Context-enhanced architectures based on Transformer backbones, including Swin Transformer [6] and Deformable DETR [7], provide strong feature modeling but incur costs that are prohibitive for high-resolution UAV tiles and are not tailored to long crack traces. Long-tail loss functions, including EQLv2 [8] and Seesaw Loss [9], improve minority-class learning on generic benchmarks but do not address the coupled localization and imbalance challenges of UAV pavement data.

C. CONTRIBUTIONS

The three components of YOLOv8-PaveDet are designed to address these needs together. GhostConv reduces redundant computation over the homogeneous pavement textures that dominate UAV pavement imagery. C2f_ConvFormer provides long-range spatial aggregation for continuous crack patterns. LSCD enforces scale-consistent multi-level prediction to handle the pronounced size variation induced by UAV altitude changes and perspective distortion.

This work makes three contributions:

- We present YOLOv8-PaveDet, a task-oriented extension of YOLOv8n combining GhostConv for efficient backbone computation, C2f_ConvFormer for long-range spatial aggregation over elongated crack structures, and LSCD for scale-consistent multi-level prediction.
- We introduce a hybrid classification loss coupling IoU-aware SlideLoss with class-weighted BCE loss, providing explicit rebalancing for minority distress classes.
- We show that GhostConv and LSCD are most effective when paired with C2f_ConvFormer, since contextual aggregation preserves thin-crack responses while the compression modules reduce model size. Without this pairing, compression-only combinations can reduce detection accuracy below the baseline.

II. RELATED WORK

A. CONTEXTUAL FEATURE ENHANCEMENT AND CROSS-SCALE FUSION

Pavement distress detection in UAV imagery is dominated by strong scale variation, texture clutter, and local semantic ambiguity, where visually similar background patterns can be confused with true defects. Recent progress therefore emphasizes strengthening contextual reasoning and cross-scale feature interaction. Hierarchical Transformer backbones such as Swin Transformer build multi-resolution representations with efficient windowed self-attention, enabling robust multi-scale feature extraction for dense prediction tasks [6], but their computational cost scales poorly with input resolution, making them impractical for high-resolution UAV tiles. Deformable DETR introduces sparse, content-adaptive attention that concentrates computation on informative regions, improving cross-scale aggregation while reducing the cost of global attention [7], yet its multi-scale attention mechanism is not tailored to long crack patterns.

Beyond backbone design, feature fusion has been enhanced through attention-driven heads that unify scale, spatial, and channel interactions. DyHead formulates detection heads as a dynamic aggregation problem with multiple attention operators, improving the quality of fused features across pyramid levels [10]. More recently, InternImage shows that deformable convolution can serve as a strong mechanism for modeling long-range dependencies and geometric variability, complementing Transformer-style context modeling in challenging scenes [11]. Related evidence from street-level recognition also supports explicit cross-scale

selection: the Cross-Scale Feature Selection Transformer adaptively selects informative scale cues under cluttered street-level imagery [12]. However, these context-enhanced architectures incur substantial computational overhead for high-resolution UAV tiles, and their attention mechanisms do not explicitly target crack continuity. These advances motivate a lightweight detector in which context aggregation and cross-scale feature sharing work together. Beyond feature representation, a second challenge in UAV pavement detection is the severe class imbalance inherent in real-world distress datasets.

B. CLASS IMBALANCE AND LOSS REWEIGHTING METHODS

Long-tailed class distributions are a persistent obstacle in real-world detection, where rare categories receive insufficient gradient signal and are easily suppressed by frequent classes. Class imbalance is commonly addressed through resampling or reweighting. While resampling can adjust the effective training distribution, it may introduce duplicated samples or discard informative majority instances; consequently, loss-level reweighting is often preferred due to its simplicity and compatibility with standard detectors. Equalization Loss v2 (EQLv2) explicitly addresses gradient imbalance by applying a gradient-guided reweighting mechanism that balances positive/negative contributions on a per-class basis, substantially improving rare-category learning in long-tailed benchmarks [8].

Several recent losses further refine this idea via adaptive suppression or dynamic competition between classes. Adaptive Class Suppression Loss reduces dominance from head classes by suppressing their gradients in a class-adaptive manner, improving tail-class optimization without requiring a multi-stage pipeline [13]. Seesaw Loss mitigates the suppression of rare classes by dynamically modulating negative gradients based on relative sample frequencies, which is effective under severe long-tail distributions [9]. Complementarily, Varifocal Loss uses IoU-aware soft targets to couple classification confidence with localization quality, improving the learning signal when positive samples are scarce [14]. Weight-Guided (WG) loss further leverages classifier weights as auxiliary prototypes for unobserved classes within a mini-batch, alleviating the “few classes per batch” issue that disproportionately harms tail categories [15]. In contrast to prior long-tail losses developed for generic benchmarks, we focus on a pragmatic hybrid objective for UAV pavement distress data, combining IoU-aware supervision with explicit class-level reweighting to better balance rare and frequent defects.

Despite these advances, a clear gap remains. Prior work leaves three needs insufficiently addressed for UAV pavement detection: efficient computation on high-resolution tiles, stronger context modeling for thin crack patterns, and better recovery of rare distress classes during training. YOLOv8-PaveDet addresses these needs within a lightweight YOLOv8n-based detector.

III. PROPOSED METHODS

A. OVERALL ARCHITECTURE

YOLOv8 is a one-stage, anchor-free detector that follows a standard four-stage design consisting of an input pre-processing module, a backbone, a neck, and a detection head [16]. The baseline employs multi-scale feature aggregation inspired by the Feature Pyramid Network (FPN) [17] and Path Aggregation Network (PAN) [18], enabling efficient fusion of low-level spatial cues and high-level semantics for objects with substantial scale variation.

We adopt YOLOv8n as the baseline because, among nano-scale detectors evaluated on UAV-PDD (Table 1), it achieves the strongest detection performance, outperforming YOLOv9t, YOLOv10n, YOLOv11n, and YOLOv12n. Its modular C2f backbone and anchor-free detection head are also directly compatible with the three proposed modifications, and the Ultralytics training framework provides a stable, reproducible training pipeline suited to controlled ablation studies.

Given an input image, the backbone extracts a hierarchy of feature maps with increasing receptive fields; the neck aggregates them into a feature pyramid to support multi-scale prediction; and the anchor-free head performs classification and bounding-box regression on each pyramid level, followed by non-maximum suppression to remove redundant detections.

Building on YOLOv8n, we construct **YOLOv8-PaveDet** as a task-oriented detector variant that targets the key characteristics of pavement distress in UAV imagery, namely thin structures, low contrast, background clutter, and fragmented crack patterns. As illustrated in Fig. 1, YOLOv8-PaveDet introduces three focused modifications: (i) GhostConv is integrated into the backbone to reduce redundant feature computation and improve efficiency; (ii) a ConvFormer-enhanced C2f block is employed to strengthen long-range dependency modeling and encourage more continuous representations of elongated cracks; and (iii) a Lightweight Shared-Channel Detection head (LSCD) is adopted to share parameters across pyramid levels, promoting scale-consistent semantics while further reducing model complexity.

Together, these modules allow feature efficiency, context aggregation, and scale consistency to reinforce one another in high-resolution UAV road scenes.

B. GHOST CONVOLUTION FOR EFFICIENT FEATURE REPRESENTATION

UAV pavement imagery contains large homogeneous regions and repetitive asphalt textures, which often yield spatial and channel-wise redundancy in intermediate convolutional features. When using standard convolutions, many output channels become highly correlated, resulting in inefficient parameter utilization and unnecessary computation. To improve efficiency without reducing feature width, we replace the standard convolutions in the C2f bottleneck modules of the backbone with Ghost Convolution

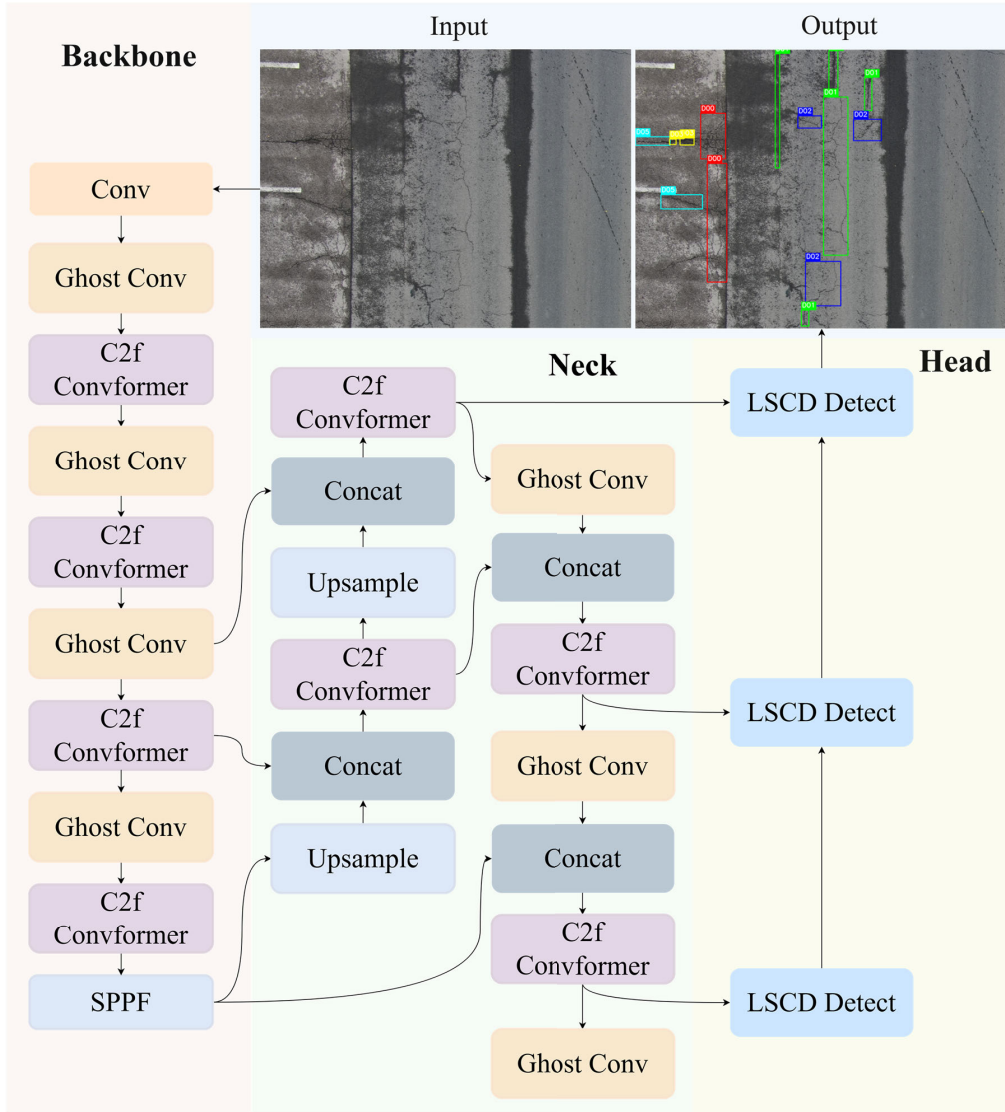


FIGURE 1. Overall architecture of the proposed YOLOv8-PaveDet framework.

(GhostConv), following the Ghost module in GhostNet [19]. A conceptual comparison between a conventional convolution and GhostConv is shown in Fig. 2.

Let $X \in \mathbb{R}^{H \times W \times C}$ denote an input feature map. A standard convolution produces

$$Y = X * f, \quad (1)$$

where $f \in \mathbb{R}^{k \times k \times C \times N}$ and N is the number of output channels. GhostConv achieves the same output width using a two-stage generation process. First, a *primary* convolution generates a compact set of m intrinsic feature maps ($m \ll N$),

$$Y' = X * f', \quad f' \in \mathbb{R}^{k \times k \times C \times m}. \quad (2)$$

Then, the remaining channels are synthesized by applying inexpensive linear operations to each intrinsic feature map:

$$Y_{i,j} = \Phi_j(Y'_i), \quad i = 1, \dots, m, \quad j = 1, \dots, s, \quad (3)$$

where $\Phi_j(\cdot)$ denotes a cheap transformation (e.g., depthwise convolution), and s is the number of derived features per intrinsic map. The final output is obtained by concatenation, yielding $N \approx m \cdot s$ channels.

When $\Phi(\cdot)$ is implemented using depthwise convolutions with kernel size k_g , the computational cost at spatial resolution $H' \times W'$ is approximately

$$\begin{aligned} \text{FLOPs}_{\text{std}} &\approx H'W' \cdot C \cdot k^2 \cdot N, \\ \text{FLOPs}_{\text{ghost}} &\approx H'W' \cdot (C \cdot k^2 \cdot m \\ &\quad + m \cdot (s - 1) \cdot k_g^2). \end{aligned} \quad (4)$$

Since $m \ll N$ and k_g is small, GhostConv substantially reduces computation while preserving output width.

In UAV pavement scenes, high-resolution imagery creates competing demands: broad coverage requires large image tiles while fine crack cues depend on narrow spatial detail.

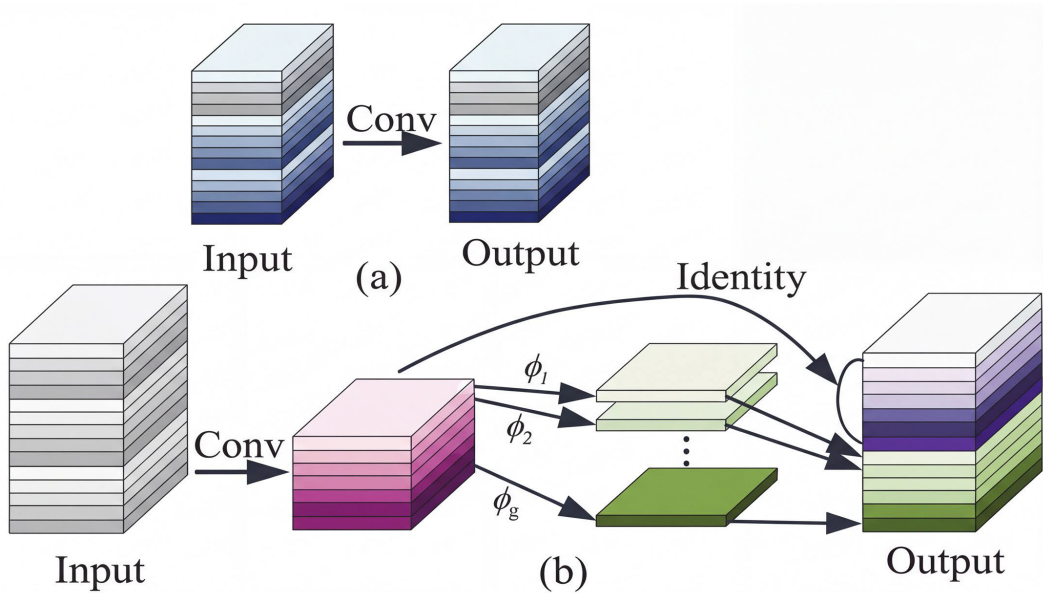


FIGURE 2. Comparison between ordinary convolution and GhostConv: (a) ordinary convolution; (b) GhostConv with intrinsic features and cheap transformations.

GhostConv reduces the computational cost of texture-heavy backbone stages without narrowing the feature channels through which crack-sensitive activations propagate, freeing processing capacity for the context-enhanced aggregation that follows in later stages of the network. The efficiency–accuracy trade-off of this substitution is quantified in Section IV-C1.

C. CONTEXT-ENHANCED FEATURE AGGREGATION VIA C2f_ConvFormer

Pavement cracks are long, thin structures whose continuity is frequently disrupted in UAV imagery by shadows, markings, and heterogeneous surface textures. With limited receptive fields, purely local convolutions can over-emphasize ambiguous local patterns and fragment the response to a single continuous crack. To strengthen spatial aggregation while maintaining efficiency, we redesign the bottleneck inside YOLOv8’s C2f module using a MetaFormer-style block, termed C2f_ConvFormer.

In a standard Transformer block, the token mixer is multi-head self-attention. Each spatial position attends to all others, computing pairwise interactions that scale quadratically with the number of tokens. For a feature map of size $H \times W$, this means $\mathcal{O}((HW)^2)$ operations, prohibitive for the high-resolution tiles used in UAV pavement inspection. C2f_ConvFormer replaces self-attention with a large-kernel depthwise separable convolution as the token mixer. Instead of global pairwise interactions, it aggregates spatial context within a fixed local neighborhood at $\mathcal{O}(HW)$ cost, which is sufficient to capture long and directionally continuous crack traces.

MetaFormer abstracts a feature block as a composition of *token mixing* (spatial interaction) and *channel mixing*

within a residual framework [20]. Instead of self-attention, which incurs quadratic complexity in the number of spatial tokens and becomes prohibitive for high-resolution UAV tiles, we instantiate token mixing using large-kernel separable convolutions, which preserve convolutional inductive bias and maintain linear complexity in the number of pixels. Concretely, we replace the original C2f bottleneck with a MetaFormer block whose token mixer is an inverted separable convolution [21].

Let $\mathbf{F} \in \mathbb{R}^{H \times W \times C}$ be the input feature map. The block contains two residual stages. The first stage performs spatial (token) mixing:

$$\mathbf{F}_1 = \mathbf{F} + \text{SepConv}(\text{LN}(\mathbf{F})), \quad (5)$$

where $\text{LN}(\cdot)$ denotes Layer Normalization and $\text{SepConv}(\cdot)$ is an inverted separable convolution. Specifically, channels are expanded by a ratio r (we use $r = 2$) through a pointwise projection, activated by StarReLU $\sigma(x) = s \cdot \text{ReLU}(x)^2 + b$, where s and b are learnable scalar parameters initialized to 1 and 0, respectively, following [20], processed by a 7×7 depthwise convolution to enlarge the effective receptive field, and projected back to C channels with a second pointwise convolution.

The second stage performs channel mixing via a lightweight MLP:

$$\mathbf{F}_{\text{out}} = \mathbf{F}_1 + \text{MLP}(\text{LN}(\mathbf{F}_1)), \quad (6)$$

where the MLP uses the same StarReLU nonlinearity to strengthen channel interactions with minimal overhead.

Embedding this MetaFormer block into the C2f split-transform–concat structure yields C2f_ConvFormer, which increases the effective receptive field through large-kernel depthwise convolutions while preserving the efficiency

characteristics of YOLOv8. Compared with heavier attention modules, this design avoids quadratic complexity and acts as a compensating mechanism for the representation loss that lightweight detectors can suffer in thin-crack scenes. It improves the continuity of crack-related responses and reduces fragmentation under cluttered backgrounds and non-uniform illumination.

D. SCALE-CONSISTENT DETECTION VIA LIGHTWEIGHT SHARED-CHANNEL HEAD

UAV-based pavement inspection exhibits pronounced scale variation due to altitude changes, perspective distortion, and the elongated geometry of crack-like targets. In standard multi-scale detection heads, each pyramid level typically uses an independent convolutional feature extractor. This design introduces parameter redundancy and can lead to *scale-wise semantic drift*, where visually similar distress patterns are mapped through different transformations at different resolutions.

To reduce redundancy while enforcing scale-consistent feature extraction, we adopt LSCD [22]. Let $\{\mathbf{P}_l\}_{l \in \{3,4,5\}}$ denote the neck outputs at different pyramid levels, where channel dimensions may differ across levels (C_l). LSCD first aligns channels into a unified hidden space using a 1×1 projection followed by Group Normalization:

$$\mathbf{F}_l^{(0)} = \text{GN}(\mathbf{W}_l^{1 \times 1} \star \mathbf{P}_l), \quad (7)$$

where $\mathbf{W}_l^{1 \times 1} \in \mathbb{R}^{d \times C_l \times 1 \times 1}$ and $d = 256$ follows the original LSCD configuration [22].

A shared feature extraction module is then applied identically to all pyramid levels:

$$\mathbf{F}_l = \text{GN}(\mathbf{W}_2 \star \text{GN}(\mathbf{W}_1 \star \mathbf{F}_l^{(0)})), \quad (8)$$

where $\mathbf{W}_1, \mathbf{W}_2 \in \mathbb{R}^{d \times d \times 3 \times 3}$ are *shared across all l*. Compared with independent per-level heads, sharing these dominant 3×3 convolutions reduces the parameter cost of the feature extraction stage by approximately 66.7%, while ensuring that features at different resolutions undergo consistent transformations.

Finally, LSCD uses lightweight decoupled prediction branches for localization and classification:

$$\mathbf{B}_l = \alpha_l \cdot (\mathbf{W}_{\text{box}} \star \mathbf{F}_l), \quad \mathbf{C}_l = \mathbf{W}_{\text{cls}} \star \mathbf{F}_l, \quad (9)$$

where $\mathbf{B}_l$ represents distribution-based box regression (trained with Distribution Focal Loss (DFL) [23]), $\mathbf{C}_l$ denotes class logits, and α_l is a learnable scale factor to compensate for level-specific regression bias.

We employ Group Normalization [24] throughout the head due to its stability under small-batch training and tiled inference, which are common in UAV pipelines. Within this architecture, LSCD serves as a cross-scale coherence mechanism that preserves YOLOv8's anchor-free paradigm while improving scale consistency and parameter efficiency.

E. HYBRID LOSS FOR LONG-TAILED PAVEMENT DISTRESS LEARNING

UAV pavement distress datasets typically follow a long-tailed distribution: frequent categories dominate the optimization signal, while rare but safety-relevant defects such as potholes contribute too few effective gradients and remain under-optimized. The problem is amplified in pavement scenes where targets are small, thin, and low-contrast, making classification confidence tightly coupled with localization quality. IoU-aware reweighting helps with ambiguous samples yet remains blind to class-frequency imbalance; we therefore adopt a hybrid objective that combines IoU-aware SlideLoss with explicit class-wise reweighting, improving minority-class learning without sacrificing localization sensitivity.

1) IOU-AWARE SLIDELOSS BASELINE

SlideLoss [25] is an intersection over union (IoU)-aware variant of binary cross-entropy (BCE) that scales the loss by a weight determined from the predicted IoU. Given a predicted score $p \in [0, 1]$, binary label $y \in \{0, 1\}$, and IoU value $\text{IoU} \in [0, 1]$, it is defined as

$$\mathcal{L}_{\text{slide}}(p, y) = -\omega(\text{IoU})[y \log(p) + (1 - y) \log(1 - p)], \quad (10)$$

where $\omega(\text{IoU})$ is a piecewise function:

$$\omega(\text{IoU}) = \begin{cases} 1.0, & \text{IoU} \leq \tau - 0.1, \\ \exp(1 - \tau), & \tau - 0.1 < \text{IoU} < \tau, \\ \exp(-(1 - \text{IoU})), & \text{IoU} \geq \tau, \end{cases} \quad (11)$$

and τ is a dynamic threshold (typically the running mean IoU during training). SlideLoss emphasizes borderline predictions and encourages refinement, but it remains agnostic to class frequency and may still bias learning toward head classes under severe imbalance.

2) PROPOSED HYBRID OBJECTIVE WITH CLASS-WISE REWEIGHTING

To explicitly rebalance gradients across distress categories, we combine SlideLoss with a class-weighted BCE term:

$$\mathcal{L}_{\text{hybrid}} = \lambda \mathcal{L}_{\text{slide}} + (1 - \lambda) \mathcal{L}_{\text{wBCE}}, \quad (12)$$

where $\lambda \in [0, 1]$ controls the trade-off. The class-weighted term is

$$\mathcal{L}_{\text{wBCE}}(p, y, c) = -w_c[y \log(p) + (1 - y) \log(1 - p)], \quad (13)$$

with class index c and a predefined class weight w_c . Larger w_c amplifies the gradient contribution of minority classes, counteracting the suppression effect caused by dominant categories. In this way, SlideLoss continues to provide IoU-sensitive supervision for hard and poorly-aligned instances, while the class-wise term ensures that rare defects remain competitive during optimization.

3) HYBRID-LOSS CONFIGURATIONS

We evaluate three configurations by varying λ and the class-weight magnitude:

- **Conservative** ($\lambda = 0.75$): $\mathbf{w} = [1.0, 1.5, 1.0, 2.5, 0.8, 1.5]$.
- **Balanced** ($\lambda = 0.65$): $\mathbf{w} = [1.2, 2.0, 1.2, 3.5, 0.9, 2.0]$.
- **Aggressive** ($\lambda = 0.55$): $\mathbf{w} = [1.5, 2.5, 1.5, 5.0, 1.0, 2.5]$.

The six elements of $\mathbf{w}$ correspond to the class order [AC, LC, Oblique Crack, Pothole, Repair, TC] as listed in Table 5. The Repair class receives a weight below 1.0 in the Conservative configuration ($w_{\text{Repair}} = 0.8$) because it already achieves relatively high AP under the baseline loss; down-weighting prevents it from dominating the gradient signal and redirects capacity toward harder categories.

These settings probe the trade-off between IoU-sensitive optimization and class-frequency compensation. Increasing $(1 - \lambda)$ strengthens the explicit class-balancing term, while larger w_c amplifies gradients for underrepresented categories. We assign the largest weight to *pothole* because it is the rarest and most safety-relevant class in the dataset. The three weight sets were designed based on the class frequency distribution of UAV-PDD, with weights proportional to the inverse square root of class frequency and scaled to three operating regimes. The specific values were validated on the held-out validation set to ensure stable and interpretable precision–recall trade-offs across configurations. The **Conservative** setting keeps SlideLoss dominant to preserve localization refinement and limit over-activation on noisy textures; the **Aggressive** setting prioritizes minority recovery under severe imbalance; and **Balanced** occupies the middle ground between localization refinement and class-frequency compensation. Together, these settings define practical reweighting regimes for comparing coverage, precision, and localization behavior. The weights are applied consistently across feature pyramid levels. During training, positives are assigned by the Task-Aligned Assigner [26], and the hybrid loss is averaged over the matched samples in each mini-batch.

IV. EXPERIMENTS

A. EXPERIMENTAL SETUP

1) DATASET AND DATA SPLIT

The UAV-PDD dataset [27] is a publicly available collection of high-resolution aerial images captured by UAVs over urban roads, specifically designed for pavement distress detection. It contains 2,440 images with a resolution of 5184×3888 pixels, collected at an approximate flight altitude of 30 meters. The dataset covers diverse road types, surface conditions, and environmental factors, including varying illumination, shadows, traffic markings, and partial occlusions by vehicles.

The distribution of the six annotated distress categories used in our experiments—longitudinal cracks (LC),

transverse cracks (TC), alligator cracks (AC), oblique cracks (OC), potholes (PH), and repair—is summarized in Fig. 3. The dataset exhibits severe class imbalance, with pothole annotations accounting for approximately 1.7% of all labeled bounding boxes. Each image is annotated with bounding-box labels following the official UAV-PDD annotation format and official data split.

Fig. 4 shows representative samples from UAV-PDD to illustrate the diversity of road types and distress patterns.

High-resolution images are cropped into smaller tiles for efficient model training and inference, while maintaining sufficient spatial context to detect thin and elongated crack structures.

2) EVALUATION METRICS

We evaluate detection performance and computational efficiency using standard metrics. For accuracy, we report Precision (P), Recall (R), F1-score, and mAP. Precision measures the proportion of predicted positives that are correct, while Recall measures the proportion of ground-truth instances that are successfully detected:

$$P = \frac{TP}{TP + FP} \quad (14)$$

$$R = \frac{TP}{TP + FN} \quad (15)$$

where TP , FP , and FN denote true positives, false positives, and false negatives, respectively. The F1-score is the harmonic mean of Precision and Recall:

$$F1 = \frac{2PR}{P + R} \quad (16)$$

We follow the COCO-style protocol for localization-aware evaluation. Specifically, mAP@0.5 denotes the mean (over classes) of Average Precision computed at an intersection over union (IoU) threshold of 0.5. We also report mAP@0.5:0.95, which averages AP over IoU thresholds from 0.5 to 0.95 with a step size of 0.05. Unless otherwise stated, we use mAP@0.5 and mAP@0.5:0.95 as the primary overall metrics; for category-wise analysis we report AP@0.5 (AP50), because thin crack annotations are sensitive to small spatial misalignment and AP50 provides a more stable indicator of detection completeness under UAV imagery.

In addition, model complexity is characterized by the number of parameters (Params, in millions) and the computational cost (GFLOPs) measured at the evaluation input resolution. Inference speed is reported as end-to-end throughput in frames per second (FPS) on a single GPU, measured under the same evaluation pipeline, including preprocessing, the model forward pass, and post-processing (non-maximum suppression).

3) IMPLEMENTATION DETAILS

All models are implemented in PyTorch and trained on a workstation running Ubuntu 22.04. Experiments are conducted with PyTorch 2.8.0 and CUDA 12.8, using a

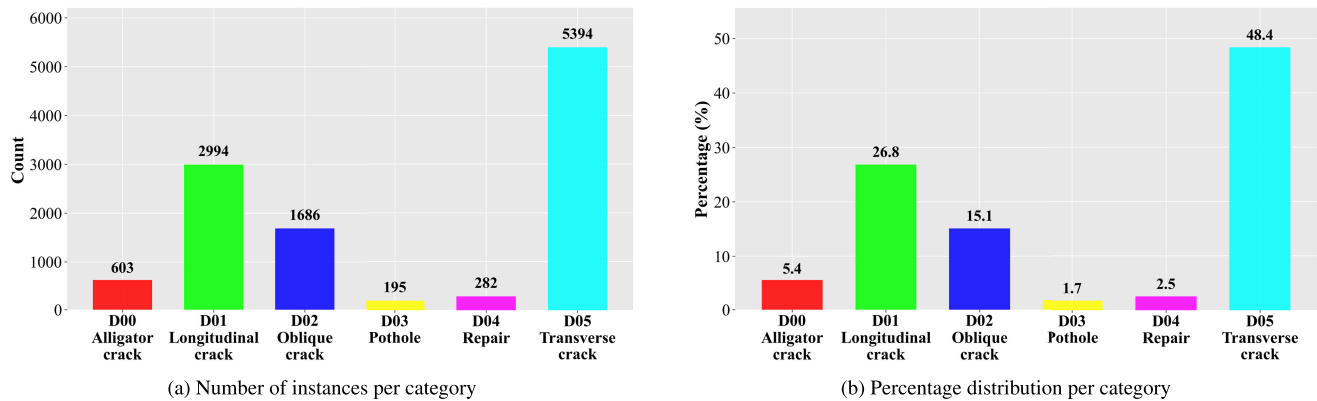


FIGURE 3. Comparison of the distribution of pavement distress categories in the UAV-PDD dataset. (a) Absolute instance count per category. (b) Relative percentage per category.

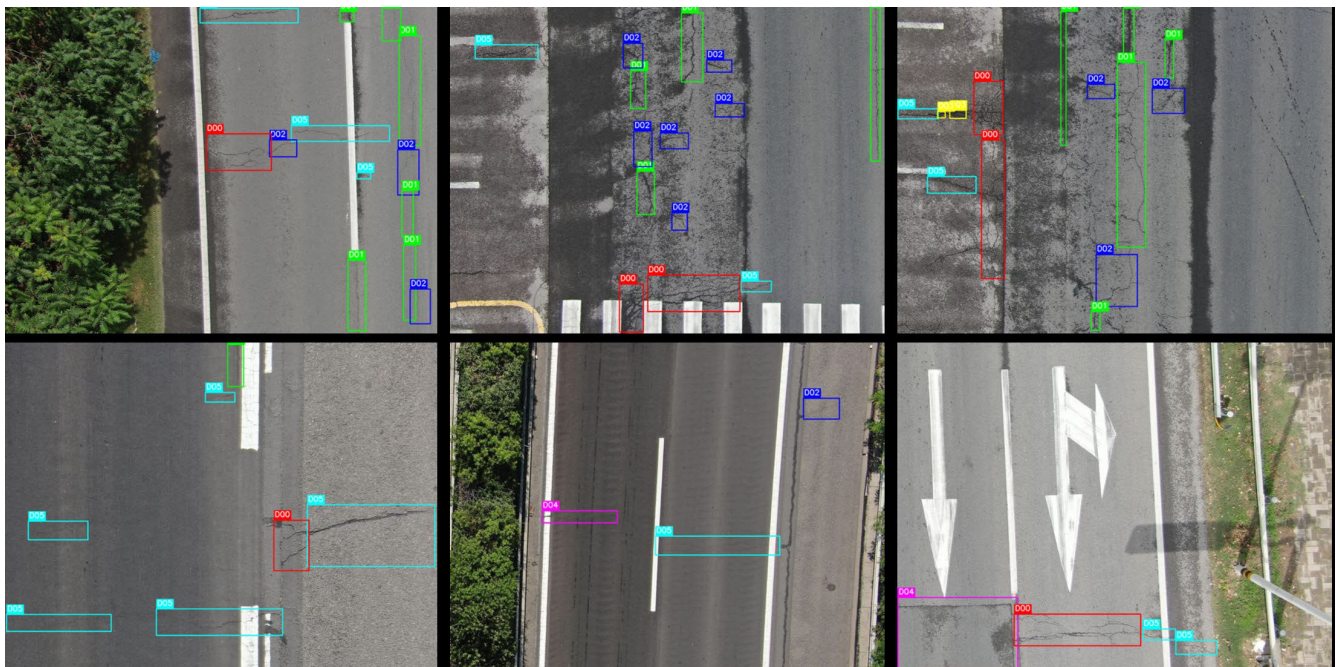


FIGURE 4. Representative sample images from the UAV-PDD dataset, highlighting various pavement distress types, illumination variations, and urban scene complexities.

single NVIDIA RTX 5090 GPU with 32 GB memory. Unless otherwise specified, all baselines and ablation variants are trained for 300 epochs with a batch size of 64, using identical data splits and training configurations to ensure a controlled comparison. Inference is performed with the same post-processing settings across methods. Unless otherwise specified, models are trained with the default YOLOv8 detection objective, including an IoU-based box regression loss with DFL and a binary cross-entropy classification loss. The FPS values reported later provide desktop-GPU reference throughput under a common evaluation pipeline. For reference, the corresponding per-tile latency can be obtained as 1000/FPS (ms). Since FPS is hardware- and implementation-dependent, Params and GFLOPs are also reported as more hardware-agnostic indicators of efficiency.

FPS is measured on individual tiles; processing a full 5184×3888 image requires multiple tiles, so full-image throughput is lower and depends on tiling stride, overlap settings, and post-processing overhead.

B. OVERALL PERFORMANCE COMPARISON

The baseline YOLOv8n reaches mAP@0.5 of 62.9%, mAP@0.5:0.95 of 32.4%, Precision of 61.0%, Recall of 66.0%, and F1-score of 62.6% (Table 1), providing the starting point against which the proposed modifications are measured.

YOLOv8-PaveDet improves across all reported metrics: mAP@0.5 increases from 62.9% to 73.4% and mAP@0.5:0.95 from 32.4% to 42.5%. Precision rises from 61.0% to 71.4% while Recall remains comparable

TABLE 1. Overall performance comparison of YOLOv8-PaveDet with the baseline YOLOv8n and other lightweight detectors on the UAV-PDD dataset.

Model	Precision (%)	Recall (%)	F1-score (%)	mAP@0.5 (%)	mAP@0.5:0.95 (%)
RT-DETR	67.8	50.6	56.9	50.1	21.3
YOLOv9t	47.6	40.6	42.6	43.1	17.1
YOLOv10n	73.4	50.6	58.1	61.3	28.0
YOLOv11n	68.3	52.9	58.6	57.7	26.7
YOLOv12n	62.7	51.4	54.4	51.6	21.5
YOLOv8n (Baseline)	61.0	66.0	62.6	62.9	32.4
YOLOv8-PaveDet (Ours)	71.4	67.1	68.5	73.4	42.5

(66.0% $\rightarrow$ 67.1%), lifting F1-score from 62.6% to 68.5%. Together these gains confirm that the proposed design improves detection quality and localization without enlarging the model. Efficiency metrics for the proposed architecture (Params, GFLOPs, FPS) are reported in the ablation study (Tables 2–3), where the full model reduces parameters from 3.01M to 1.83M and GFLOPs from 4.10 to 2.78 relative to the baseline. The effect of the hybrid loss on detection performance is reported in Section IV-C.

To our knowledge, no prior crack-specific or UAV pavement detection method has published results on the UAV-PDD benchmark. The comparison is therefore conducted against general-purpose lightweight detectors evaluated under the same protocol, including RT-DETR, YOLOv9t, YOLOv10n, YOLOv11n, and YOLOv12n. YOLOv8-PaveDet achieves the best detection performance among the compared models, obtaining the highest mAP@0.5 (73.4%) and mAP@0.5:0.95 (42.5%). Relative to the strongest alternative, YOLOv10n, mAP@0.5 improves from 61.3% to 73.4% and mAP@0.5:0.95 from 28.0% to 42.5%. Although YOLOv10n shows slightly higher Precision (73.4% vs. 71.4%), YOLOv8-PaveDet achieves higher Recall (67.1% vs. 50.6%) and the best F1-score (68.5%), indicating a more favorable balance between detection completeness and false positives on UAV-PDD.

C. ABLATION STUDY

To quantify the contribution of each proposed component and examine whether their effects are complementary, we conduct ablation experiments in two stages: (i) inserting one module at a time into the YOLOv8n baseline and (ii) evaluating pairwise combinations and the full model. All variants are trained and evaluated under the same protocol. We report overall detection performance using mAP@0.5 and mAP@0.5:0.95, together with efficiency indicators including model complexity (Params, GFLOPs) and inference throughput (FPS). For clarity, all accuracy metrics are presented in percentage points.

1) IMPACT OF INDIVIDUAL MODULES

We first isolate the effect of each module by inserting it individually into the baseline (Table 2). All three modules improve average precision over YOLOv8n, but they do so with different accuracy–efficiency profiles. ConvFormer gives the largest gain under the stricter mAP@0.5:0.95 metric

(32.43% $\rightarrow$ 36.27%, +3.84), indicating better localization. LSCD reduces parameters and GFLOPs while keeping similar throughput, and GhostConv also lowers model size and computation while maintaining detection gains. The main runtime cost comes from ConvFormer because it adds contextual aggregation.

2) EFFECT OF MODULE COMBINATIONS

We next evaluate module interactions (Table 3). ConvFormer+GhostConv is the strongest two-module setting, raising mAP@0.5:0.95 from 32.43% to 40.88% and mAP@0.5 from 62.94% to 69.69%. The result suggests that reducing redundant texture computation helps the context block use capacity more effectively. In contrast, GhostConv+LSCD performs below the baseline, indicating that stacking compression-oriented modules can weaken thin-crack features when no context block is added.

Integrating all three modules yields the best overall detection quality with the smallest model footprint. The full YOLOv8-PaveDet model reaches 73.40% mAP@0.5 and 42.53% mAP@0.5:0.95, with model complexity reduced to 1.83M parameters and 2.78 GFLOPs. Its FPS is lower than YOLOv8n because ConvFormer dominates runtime, while GhostConv and LSCD still reduce parameters and GFLOPs.

3) LOSS FUNCTION ABLATION

With the architecture fixed to YOLOv8-PaveDet (ConvFormer + GhostConv + LSCD), we compare SlideLoss with three hybrid-loss settings in Table 4. The only variables are λ and the class weights $\{w_c\}$. SlideLoss alone performs slightly below the default BCE objective on this dataset (72.06 vs. 73.40 mAP@0.5 for the full architecture), likely because its IoU-aware weighting is less effective without explicit class rebalancing under severe long-tail distributions.

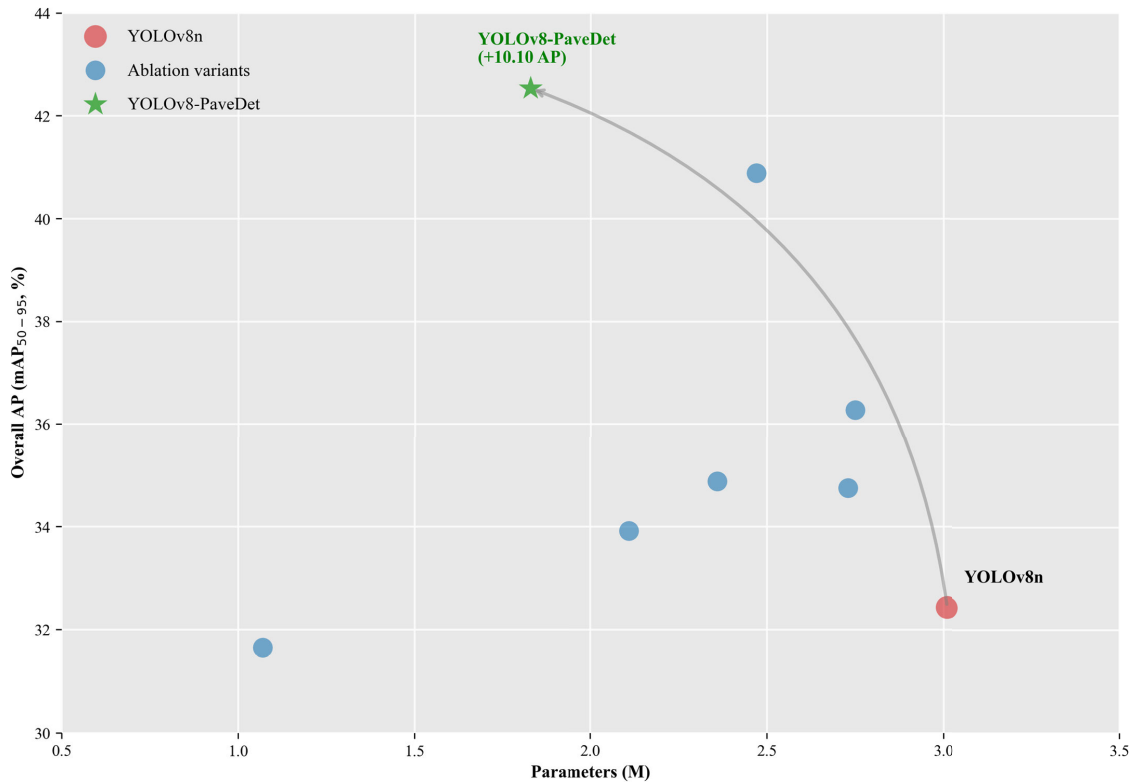
As shown in Table 4, all hybrid losses improve over SlideLoss on both mAP@0.5 and mAP@0.5:0.95, confirming the value of class-wise reweighting for UAV-PDD. The settings trade precision, recall, and localization quality: **Balanced** gives the highest Precision, **Aggressive** gives the highest mAP@0.5 and Recall, and **Conservative** gives the highest mAP@0.5:0.95. Thus, the preferred setting depends on whether the operating objective prioritizes coverage, stricter localization, or a compromise between precision and recall.

TABLE 2. Ablation results of individual modules. Accuracy metrics are in percentage points. Efficiency metrics are measured under the same input resolution and inference settings.

Method	mAP@0.5	mAP@0.5:0.95	GFLOPs	Params (M)	FPS
Baseline (YOLOv8n)	62.94	32.43	4.10	3.012	278.29
+GhostConv	67.86	34.74	3.83	2.732	259.30
+ConvFormer	67.32	36.27	3.84	2.756	146.04
+LSCD	67.68	34.87	3.31	2.367	280.07

TABLE 3. Ablation results of module combinations. Accuracy metrics are in percentage points. Efficiency metrics are measured under the same input resolution and inference settings.

Method	mAP@0.5	mAP@0.5:0.95	GFLOPs	Params (M)	FPS
Baseline (YOLOv8n)	62.94	32.43	4.10	3.012	278.29
+ConvFormer + GhostConv	69.69	40.88	3.57	2.476	138.85
+ConvFormer + LSCD	65.97	33.91	3.06	2.110	146.14
+GhostConv + LSCD	60.56	31.65	1.79	1.075	203.51
+ConvFormer + GhostConv + LSCD (Ours)	73.40	42.53	2.78	1.831	139.24

**FIGURE 5.** mAP@0.5:0.95 versus parameter count for the baseline, ablation variants, and the final model. The proposed modules progressively move the detector toward a better accuracy–efficiency regime, and the final model achieves the largest AP gain with substantially fewer parameters.

D. CATEGORY-WISE PERFORMANCE ANALYSIS

Table 5 summarizes per-class AP@0.5 (AP50) and the relative change with respect to the baseline. We emphasize AP50 rather than AP@0.5:0.95 in this analysis because thin crack annotations are highly sensitive to small spatial misalignment in UAV imagery; mAP50 provides a more stable indicator of detection completeness under such conditions.

The clearest gains concentrate on the most underrepresented category, *Pothole*. Under stronger reweighting,

performance rises sharply, and the Aggressive variant nearly doubles AP50 (39.70% → 75.81%). This behavior indicates that explicit class-level emphasis is important for recovering minority instances that remain under-optimized under the baseline loss.

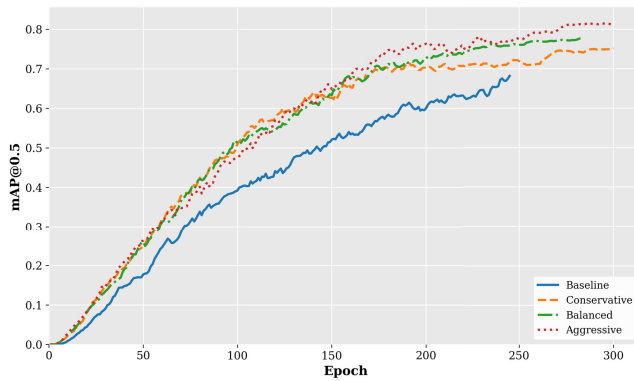
Frequent or visually dominant categories remain comparatively stable, although mild regressions appear when learning capacity is shifted toward rare classes. The effect is most visible for *Alligator Crack*, where Conservative

TABLE 4. Ablation of hybrid loss configurations on UAV-PDD with a fixed YOLOv8-PaveDet architecture (ConvFormer + GhostConv + LSCD). The baseline corresponds to SlideLoss (no class reweighting); the other rows apply the proposed hybrid loss with different λ settings.

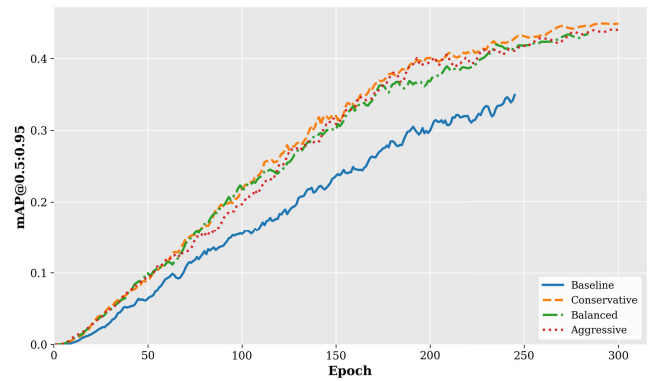
Loss Configuration	mAP@0.5 (%)	mAP@0.5:0.95 (%)	Precision (%)	Recall (%)
Baseline (SlideLoss)	72.06	39.03	70.97	69.03
Conservative ($\lambda = 0.75$)	75.71	45.54	82.41	70.38
Balanced ($\lambda = 0.65$)	78.60	44.86	85.89	68.32
Aggressive ($\lambda = 0.55$)	81.68	44.88	82.66	75.62

TABLE 5. Category-wise AP@0.5 comparison across loss configurations. All values are percentages. Δ denotes the relative improvement over the baseline SlideLoss.

Category	Baseline	Conservative		Balanced		Aggressive	
	AP50	AP50	Δ (%)	AP50	Δ (%)	AP50	Δ (%)
AC (Alligator Crack)	85.45	81.60	-4.50	82.74	-3.17	86.65	+1.41
LC (Longitudinal Crack)	81.35	86.67	+6.54	82.00	+0.80	86.69	+6.56
Oblique Crack	67.67	74.21	+9.67	71.31	+5.38	70.50	+4.18
Pothole	39.70	33.62	-15.32	61.72	+55.46	75.81	+90.94
Repair	76.34	91.91	+20.39	91.50	+19.85	86.26	+12.99
TC (Transverse Crack)	81.84	86.22	+5.35	82.34	+0.61	84.18	+2.86
Overall (mAP@0.5)	72.06	75.71	+5.06	78.60	+9.08	81.68	+13.35



(a) mAP@0.5



(b) mAP@0.5:0.95

FIGURE 6. Validation mAP curves for different loss configurations (Baseline, Conservative, Balanced, Aggressive). (a) mAP@0.5. (b) mAP@0.5:0.95.

and Balanced slightly underperform the baseline, while Aggressive remains close to neutral (+1.41%). *Repair*, by contrast, benefits strongly across all hybrid variants and reaches its highest AP50 under the Conservative setting (91.91%), which suggests that improved optimization and class-aware balancing are both helpful for this category.

The crack categories improve more moderately but still consistently. *Longitudinal* and *Transverse* cracks gain under both Conservative and Aggressive settings, while *Oblique Crack* improves most under Conservative. The pattern supports a restrained interpretation: hybrid reweighting helps thin, elongated structures, but more aggressive weighting is not uniformly optimal across all crack geometries.

E. DETECTION PERFORMANCE VISUALIZATION

Validation mAP curves for the baseline SlideLoss and the three hybrid variants reveal convergence speed, training stability, and the performance gap maintained across epochs, complementing the final numbers in Table 4.

Three complementary views are included: (i) the mAP@0.5 and mAP@0.5:0.95 evolution over training epochs, (ii) a representative false-positive pattern on the most imbalanced category (Pothole), and (iii) detection completeness on thin, elongated crack structures.

Pothole is the most underrepresented category in UAV-PDD and shows the strongest sensitivity to reweighting strength (Table 5). To illustrate a potential side effect of aggressive reweighting, we visualize a representative scene in which the Aggressive configuration produces an additional pothole detection on a visually confusing background region that is not annotated in the ground truth. This example indicates that stronger penalization can make the detector more responsive to subtle pothole-like cues, but may also increase susceptibility to texture-induced false positives in cluttered pavement surfaces.

Thin, elongated crack structures remain challenging under UAV imagery due to low contrast, cluttered surface texture, and frequent discontinuities induced by shadows and



FIGURE 7. False-positive illustration on Pothole (same scene). Compared with the ground truth, the Aggressive configuration introduces an additional pothole detection on a visually confusing region.

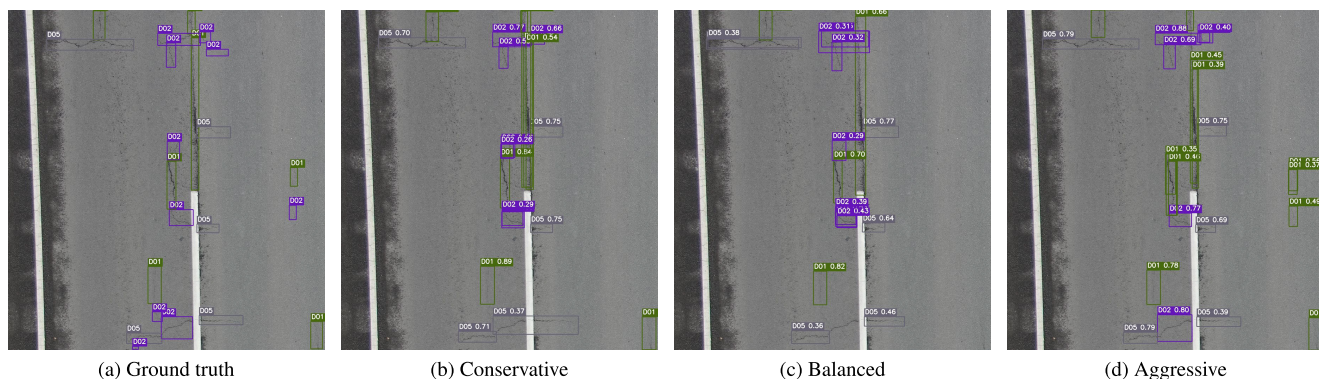


FIGURE 8. Qualitative comparison on a representative thin-crack sample. Ground truth is shown alongside predictions from Conservative, Balanced, and Aggressive hybrid losses. The Aggressive configuration reduces fragmented detections and recovers faint crack segments under texture clutter and shadow interference, improving both coverage and trace continuity.

surface wear. Predictions on such samples easily fragment into multiple short boxes, degrading detection completeness and crack trace continuity. Fig. 8 contrasts ground-truth annotations with predictions from the three hybrid configurations on a representative crack-dominant scene; the Aggressive setting produces more continuous detections and recovers faint crack segments commonly missed under weaker reweighting.

V. DISCUSSION

UAV pavement distress detection benefits most from coordinated lightweight design rather than from isolated compression or isolated reweighting. Across the ablations, the full model improves detection quality more reliably than any single efficiency-oriented modification applied on its own.

A. ACCURACY-EFFICIENCY TRADE-OFFS

The architectural ablations show that better results do not come from increasing model size. Instead, the gains arise from how feature efficiency, contextual aggregation, and cross-scale consistency interact under high-resolution

road imagery. GhostConv reduces redundant texture-dominated computation, which is valuable in UAV scenes filled with repetitive asphalt patterns; ConvFormer contributes the contextual support needed to preserve elongated crack continuity; and LSCD keeps multi-scale semantics more consistent without restoring a heavy prediction head.

GhostConv+LSCD alone underperforms the baseline, whereas ConvFormer+GhostConv produces a substantial improvement and the full model achieves the best balance across all reported metrics (Tables 2–3). The result indicates that lightweight compression cannot simply be stacked in thin-crack scenes; efficiency gains must be coupled with stronger contextual modeling so that fine structure survives the compression. This interaction is one of the clearest empirical contributions of the paper, since it yields a transferable design principle even though the individual modules are known in isolation.

Class-aware hybrid reweighting improves minority-category learning, and the preferred operating point depends on whether the priority is coverage or stricter localization. The Aggressive setting recovers many more potholes and

delivers the highest mAP@0.5, whereas the Conservative setting preserves the strongest mAP@0.5:0.95. These results frame the loss as a controllable trade-off across coverage, precision, and localization.

These gains matter in practice because the most difficult defects are also the most safety-relevant: thin cracks demand continuity-sensitive features, and rare potholes carry disproportionate risk despite their small data share. Improved detection quality therefore translates into more reliable upstream evidence for inspection workflows, independent of any downstream scoring or assessment layer.

B. DEPLOYMENT CONSIDERATIONS

YOLOv8-PaveDet provides a compact computation footprint for edge deployment. Recent benchmarks on drone-oriented edge deployment report that YOLOv8n achieves 52 FPS on the NVIDIA Jetson Orin NX at FP32 precision, rising to 65 FPS with INT8 quantization [28]. However, embedded-device throughput cannot be inferred from GFLOPs alone because operator implementations and memory-access patterns differ across platforms. Therefore, the Jetson Orin NX results are used only as deployment-oriented context, and direct TensorRT/INT8 benchmarking on representative UAV edge hardware remains future work.

C. STUDY LIMITATIONS AND SCOPE

This study is confined to the detection stage of UAV pavement inspection. All experiments are conducted on a single UAV distress dataset, so the evidence here is restricted to high-resolution aerial road scenes with the annotation protocol and class distribution provided by UAV-PDD. Broader validation across additional datasets, acquisition conditions, and pavement environments would clarify how well the same design carries to other inspection settings.

The robustness of the model under diverse environmental conditions, including varying weather (rain, overcast, direct sunlight), extreme camera angles, and night-time illumination, has not been systematically evaluated, as UAV-PDD does not provide labeled data under such conditions. The dataset includes natural variation in illumination, shadows, and surface wear, and the qualitative results suggest reasonable performance under these conditions. Dedicated robustness experiments under controlled environmental perturbations remain an important direction for future work, particularly for deployment in regions with highly variable weather.

All experiments are conducted on UAV-PDD, which covers urban road types at a specific flight altitude (~ 30 m) with a particular annotation protocol and class taxonomy. The design choices most likely to transfer to other datasets are the hybrid loss (applicable wherever class imbalance exists) and C2f_ConvFormer (applicable to any thin elongated structure detection task). The class weights in the hybrid loss are tuned to the UAV-PDD class distribution and would require re-calibration for datasets with different

imbalance ratios. Broader validation on datasets such as RDD2022 and CrackForest is planned as future work.

The task formulation is also deliberately narrow. The model predicts bounding boxes for distress categories, whereas engineering workflows often require severity estimation, extent measurement, segment-level condition indicators, and comparisons across repeated surveys. The detector outputs can be exported as georeferenced layers and aggregated into road-segment summaries, but the paper stops short of evaluating a complete engineering assessment workflow. Extending the present setting toward segment-level assessment, multi-temporal deterioration analysis, and richer condition descriptors remains a natural next step.

VI. CONCLUSION

YOLOv8-PaveDet extends YOLOv8n for UAV pavement distress detection through a coordinated combination of efficient feature extraction, context enhancement, cross-scale head sharing, and class-aware reweighting. On UAV-PDD, this design improves overall detection accuracy and minority-class sensitivity while reducing parameter count and GFLOPs relative to the baseline, which supports its use in high-resolution aerial road inspection where both detail preservation and model efficiency matter.

The full three-module combination produces the largest gains (+10.46% on mAP@0.5, +10.10% on mAP@0.5:0.95 over the baseline architecture). In contrast, GhostConv and LSCD applied without C2f_ConvFormer reduce recall below baseline, showing that efficiency-oriented compression should be paired with context-enhanced aggregation to preserve thin-crack response. Hybrid reweighting addresses a separate bottleneck in training bias; under the Aggressive setting, Pothole AP@0.5 rises from 39.70% to 75.81%, while the Conservative setting maintains the highest mAP@0.5:0.95 at 45.54%, giving practitioners a controllable trade-off between coverage and localization fidelity.

The current scope remains limited to box-level distress detection on a single UAV dataset. In practice, the detector is most directly suited to UAV road-corridor screening, where predicted boxes can be reviewed or aggregated before maintenance decisions. Field use should still be validated under different flight altitudes, weather, illumination conditions, and embedded hardware. Broader validation across additional pavement environments, together with links from detections to segment-level condition indicators, severity descriptors, repeated-survey analysis, and public infrastructure management, would extend the present results toward fuller engineering assessment pipelines.

ACKNOWLEDGMENT

We thank the editor and reviewers for their helpful suggestions, which improved the manuscript.

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